

Stereo Vision

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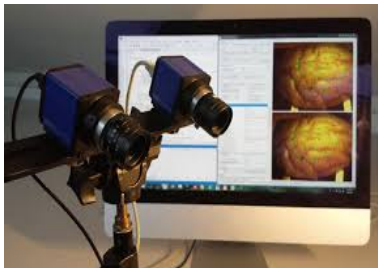
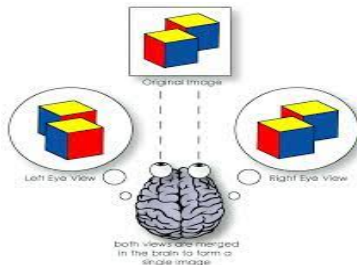
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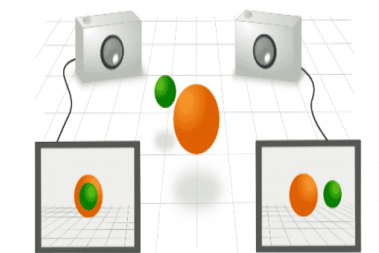


- 1 Goal of stereo vision
- 2 Visual cues for 3D
- 3 Estimating depth with Stereo when Cameras are parallel

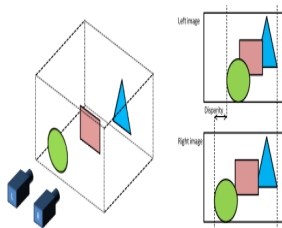
To mimic human vision system



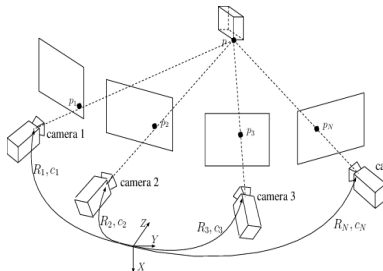
What extra information captured by two cameras, compared to one



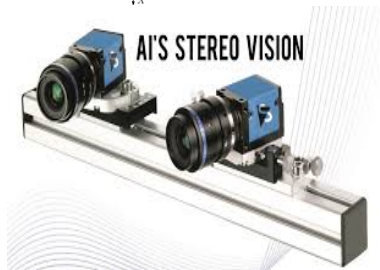
Need of Stereo Vision



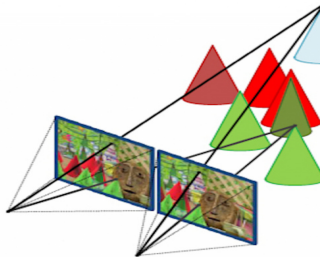
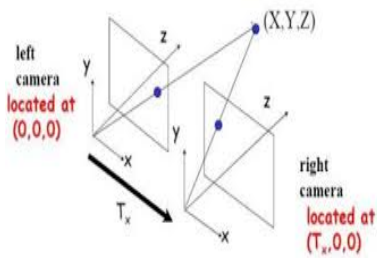
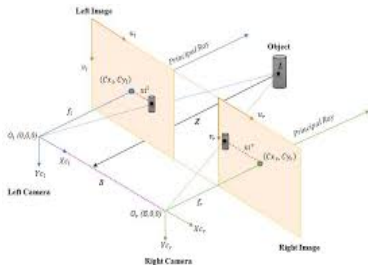
1. Goal of stereo Vision: The recovery of the 3D structure(3D points, shape etc) of a scene using two or more images of the 3D scene, each acquired from a different viewpoint in space.



2. The term binocular vision is used when two cameras are employed.



Goal of stereo vision (cont.)





When we look at image, what properties indicate the differences in depth or provide hints about object's shape?











- ▶ X = shading, texture, focus, motion, ...
- ▶ We'll focus on the motion cue



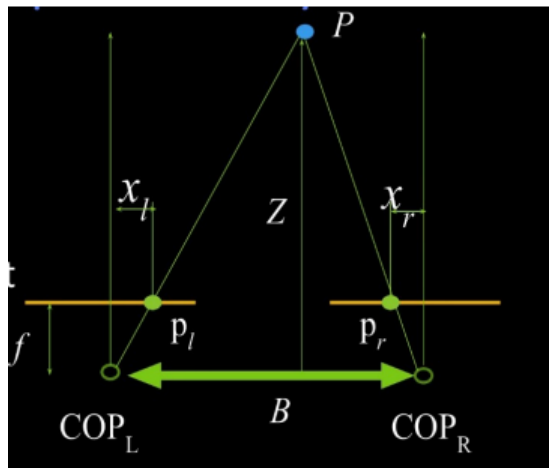
The Cameras are said to be in parallel if

- ▶ CCS of Camera 1 is aligned with WCS
- ▶ X Y and Z axes of CCS1 and CCS2 are the same
- ▶ Origin of CCS 2 is displaced from the origin of CCS 1 by $(B, 0, 0)$

To find depth from two images captured by parallel cameras

- Find the Corresponding points in left and right images, for the same object point
- Find x_l and x_r
- Compute Depth: $Z = \frac{Bf}{(x_l - x_r)}$
 - ▶ f and B will be given by stereo camera manufacturer

Geometry for a stereo system



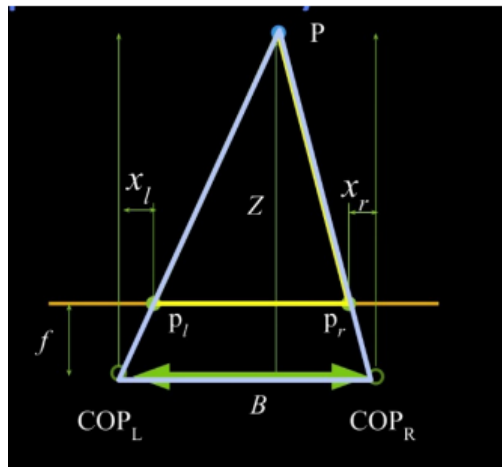


- ▶ B is baseline width, distance between centres of the cameras
- ▶ **Defn of depth:** The distance between a point p and the baseline is called as depth of p

Observe that:

- ▶ Two Image planes are coplanar
- ▶ Image planes at the front of image camera
- ▶ Origins are at the center of the planes
- ▶ Displacement (from origin) is positive in left image and negative in right image
- ▶ Z is the depth to be obtained from the two formed images.
- ▶ f is the focal length.

Geometry for a stereo system (cont.)





- ▶ Similar triangles (P_l, P, P_r) and (COP_L, P, COP_R)

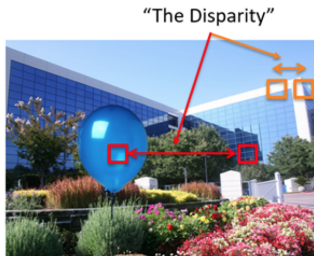
$$\frac{B - x_l + x_r}{Z - f} = \frac{B}{Z}$$
$$Z = \frac{f B}{x_l - x_r}$$

- ▶ $x_l - x_r$ is called as disparity
- ▶ Depth is inversely proportional to Disparity

Disparity map



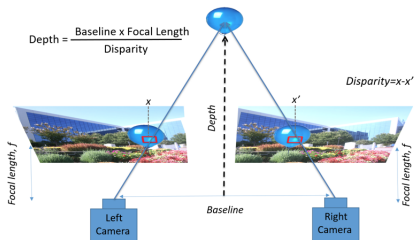
Left camera



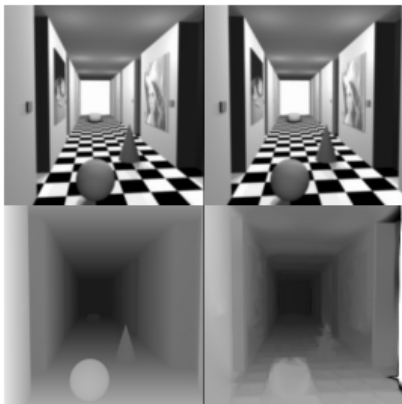
Right camera

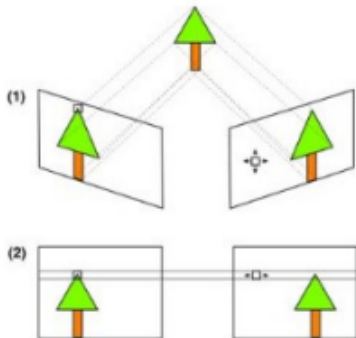


Disparity Map referenced to Right camera



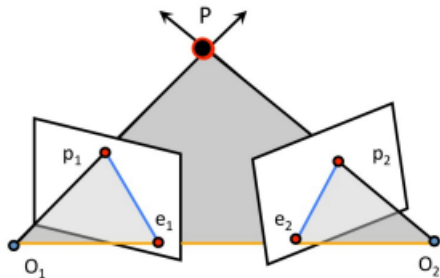
Disparity map





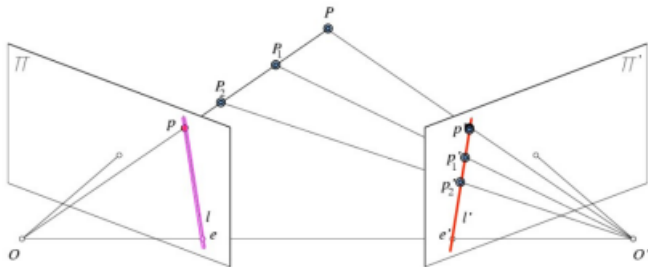
Search problem: Given an element in the left image, we search for the element in the right image. This involves two decisions:

- ▶ Which image element to match
- ▶ Which similarity measure to adopt



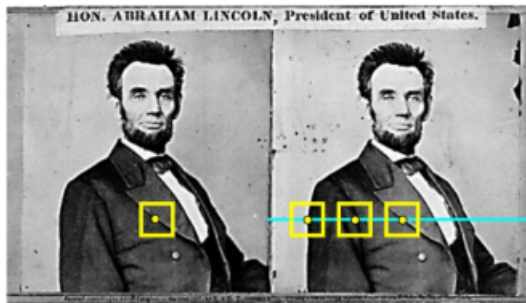
- ▶ The standard epipolar geometry setup involves two cameras observing the same 3D point P , whose projection in each of the image planes is located at p_1 and p_2 .
- ▶ e_1 and e_2 are point of intersection of image plane with base line for left and right cameras respectively
- ▶ e_1 and e_2 are called epipoles.
- ▶ Epipolar line is the intersection of an epipolar plane with the image plane

Epipolar Constraint



- ▶ Potential matches for p have to lie on the corresponding epipolar line l
- ▶ Potential matches for p' have to lie on the corresponding epipolar line l'

Why is the epipolar constraint useful?



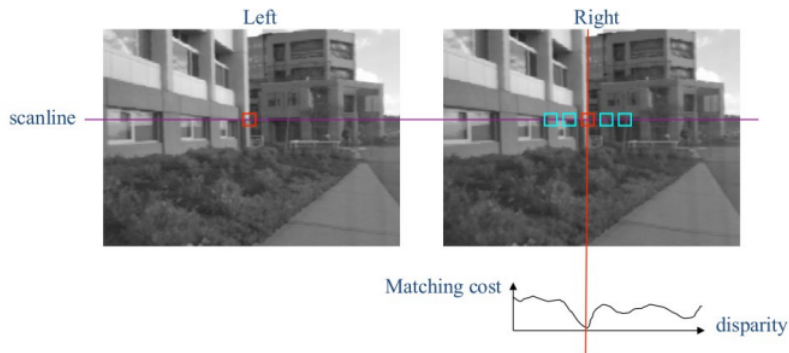
The epipolar constraint reduces the correspondence problem to a 1D search along the epipolar line.



Beyond the constraint by epipolar geometry , there are other constraints to help identify corresponding points.

- ▶ **Similarity** - Image patch from the left should match with the right
- ▶ **Uniqueness** - There is no more than one match for the pixel in right image
- ▶ **Ordering** - If pixels go a,b,c in left, they go a,b,c in right
- ▶ **Disparity Gradient is limited** - Depth doesn't change too quickly

Correspondence search with similarity constraint

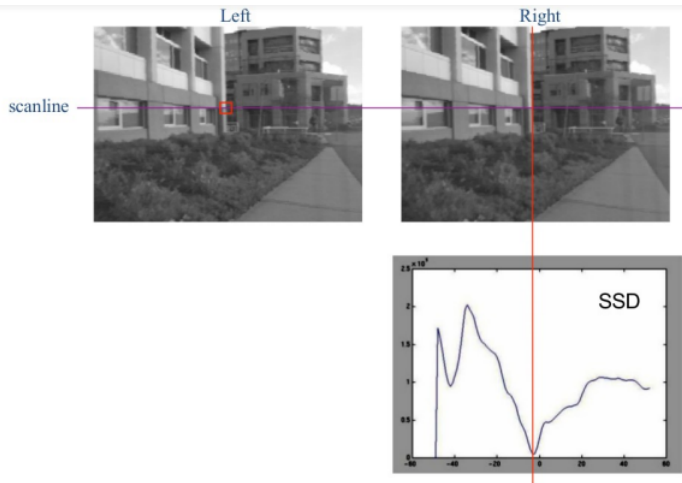


- ▶ Sum of Squares difference(SSD)
- ▶ Normalized Correlation

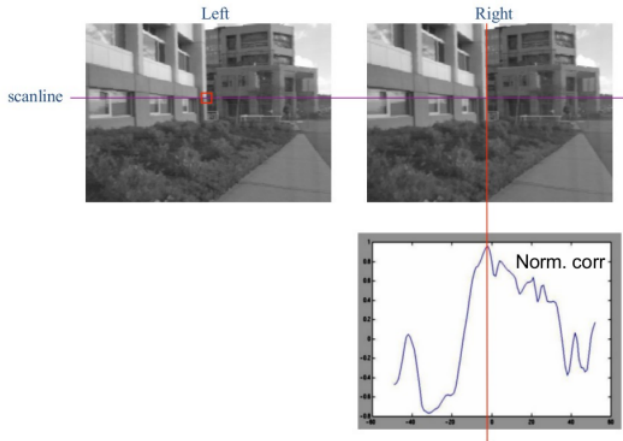
$$r_{ij} = \frac{\sum_m \sum_n [f(m+i, n+j) - \bar{f}][g(m+i, n+i) - \bar{g}]}{\sqrt{\sum_m \sum_n (f(m+i, n+j) - \bar{f})^2 \sum_m \sum_n (g(m+i, n+i) - \bar{g})^2}}$$

- ▶ Mutual Information

$$I(X; Y) = \int_y \int_x P(x, y) \log \left(\frac{P(x, y)}{P(x)P(y)} \right) dx dy$$

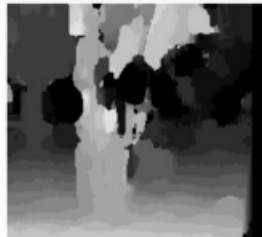


Normalized Correlation





$W = 3$



$W = 20$

- ▶ Smaller window
 - More detail
 - Less noise
- ▶ Larger window
 - Smoother disparity maps
 - Less detail

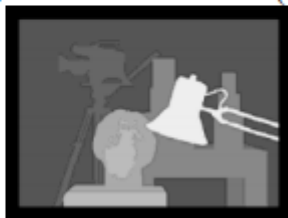
Disparity Map



(a)



(b)



(c)



- ▶ Low-contrast; Texture less image regions
- ▶ Occlusions
- ▶ Large base lines
- ▶ Camera calibration errors etc.,